

Ans Sheet: CS & AI Master Exam - Priya Nair (Roll: CS-2026-002)Q1: Backpropagation

Backpropagation is used in deep learning to train neural networks.

It consists of forward pass to compute output and loss function, and backward pass to update weights.

Weights are adjusted using gradient descent method $w = w - \eta * \text{gradient}$.

It calculates how much each node contributed to final error.

Q2: QuickSort vs MergeSort Complexity

MergeSort divides array into two halves, sorts them, and merges back.

Time complexity is $O(N \log N)$ in all cases (best, average, worst). Space complexity is $O(N)$.

QuickSort picks a pivot element and partitions array around it.

Average time complexity is $O(N \log N)$. Space complexity is $O(1)$ in-place.

Worst case time is $O(N^2)$ when array is sorted.

Q3: Database ACID Properties

ACID stands for:

- Atomicity: All operations in transaction succeed or fail together.
- Consistency: Data remains consistent before and after transaction.
- Isolation: Multiple users reading/writing database do not clash.
- Durability: Saved data is retained permanently on hard drive.

Q4: OOP Principles

1. Encapsulation: Keeping variables private in a class so outside code cannot modify directly.
2. Inheritance: Subclass inherits properties from parent class using extends keyword.
3. Polymorphism: Same function name behaving differently depending on arguments or class type.
4. Abstraction: Hiding background details from user.

Q5: RSA Public Key Cryptography

RSA uses asymmetric cryptography with a public key for encryption and a private key for decryption.

Public key is shared with everyone, while private key is kept secret by owner.

Sender encrypts message using recipient's public key. Only holder of private key can decrypt ciphertext.

Uses prime numbers p and q to create large modulus n .